Effects of maternal alcohol intake and smoking on neonatal electroencephalogram and anthropometric measurements

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Anthropometric data and computerized electroencaphalogram analysis during quiet, indeterminate, and active sleep were obtained from infants of mothers of four groups: (1) heavy drinking mothers (>2 ounces of alcohol per day); (2) nondrinking mothers; (3) smoking, nondrinking mothers; (4) nonsmoking, nondrinking mothers. Infants in groups 1 and 2 were matched as closely as possible for postconceptional age, sex, race, and socioeconomic status. Infants in groups 3 and 4 were matched similarly. Infants of alcoholic mothers had a significantly lower birth weight, length, and head circumference than those from the matched control group. Infants of smoking mothers had lower birth weights and lengths than infants of nonsmoking mothers, but head circumference was identical. Hypersynchrony of the electroencephalogram was seen only in "alcoholic" infants, and power spectral density analysis revealed that the average integrated power was significantly increased in quiet, active, and indeterminate sleep. The greatest increase in electroencephalogram power (212%) was seen in active sleep, and this analysis clearly separated 15 of 17 alcohol-exposed infants from the control infants. These data suggest that alcohol has a specific toxic effect on the fetal brain that is not linked with smoking habits. The neonatal electroencephalogram is affected even in the absence of dysmorphology and thus may be the most sensitive indicator of fetal alcohol toxicity. (AM. J. OBSTET. GYNECOL. 146:41, 1983.)

IT HAS BEEN ONLY relatively recently appreciated that maternal ingestion of alcohol may have harmful effects on the developing fetus.[1-3] We have previously described significant changes in the electroencephalogram power spectrum of infants of alcoholic mothers who did not necessarily have the gross congenital anomalies associated with the fetal alcohol syndrome.[4] Maternal smoking also has been shown to have harmful effects on fetal growth[5-7] and development.[8] Most of our alcoholic mothers also smoke, and a clear separation of the effects of alcohol from those of smoking on the neonatal electroencephalogram and morphometric data has not been previously undertaken. The present study therefore was designed to compare the electroencephalogram power spectrum analysis and morphometric data of infants of alcoholic mothers with infants of smoking, nondrinking mothers carefully matched to their own control groups.

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Subjects and methods

In this study we used data which were obtained prospectively from 70 infants born between April, 1975, and July, 1980, at the Women's Hospital, Health Sciences Centre, Winnipeg. Infants of mothers who were heavy users of alcoholic beverages (>2 ounces of absolute alcohol per day) were selected for study.[9] Control infants of mothers who were only occasional drinkers (<1 ounce of alcohol on any occasion) or abstainers were matched as closely as possible for postconceptional age, sex, race, socioeconomic status, and smoking history. Similarly, infants of smoking mothers who were nondrinkers were selected for study and matched

Table I. Maternal characteristics

Parameter	Subject: Control	Subject: Alcoholic	Subject: Control	Subject: Smoking
Age (yrs)				
16-18	1	1	3	2
19-30	14	8	14	15
31-35	2	8	1	1
Mean ± SEM	24.2 ± 1.26	27.7 ± 1.75	23.5 ± 1.05	25.1 ± 1.08
Race				
American Indian	15	15	4	4
White	2	2	14	14
Parity				
0	1	3	5	3
1-3	11	8	11	10
>4	5	6	2	5
Mean ± SEM	2.8 ± 0.63	3.4 ± 0.74	0.81 ± 0.16	1.67 ± 0.34
Smoking history				
0 (none)	6	4	18	0
1 (occasional)	4	0	0	0
2 (0.5 pack/day)	6	2	0	0
3 (1 pack/day)	1	8	0	14
4 (>1 pack/day)	0	2	0	4
Mean ± SEM	1.12 ± 0.24	2.25 ± 0.36*	0.00	3.22 ± 0.1*
Pregnancy risk factors				
Preeclampsia	3	1	1	0
Diabetes	0	0	0	0
Alcohol withdrawal	0	0	0	0
Epilepsy	0	0	0	0
Other diseases	0	0	0	3
Illicit drugs	0	0	0	0
Socioeconomic class				
0	10	16	2	3
1	4	0	2	3
2	3	1	3	5
3-5	0	0	9	5
Mean	0.12 ± 0.12*	0.59 ± 0.19	3.00 ± 0.47	2.07 ± 0.38

*$p < 0.001$; $p < 0.05$ statistical difference, mean ± SEM.

as closely as possible for postconceptional age, sex, race, and socioeconomic status with a control population of infants of nonsmoking, nondrinking mothers. Thus, four groups of infants were studied: alcohol-exposed and their control population and smoke exposed and their control population. All infants were born by vaginal delivery either under epidural anesthesia or with 50% nitrous oxide in oxygen analgesia.

Informed consent to study their infants and administer a questionnaire was obtained from the mothers on admission to the labor floor. The questionnaire pertained to maternal habits, beverage intake, smoking history, and other pertinent information such as use of drugs or other substances during the pregnancy. Information from the questionnaire was confirmed with the patient's physician or social worker. Gestational age of the baby was estimated on the basis of date of the last normal menstrual period and a Dubowitz assessment.[10] Where a discrepancy existed the Dubowitz assessment was taken as the more accurate determination of gestational age. Only infants between 37 and 42 weeks' gestational age were selected for study. Measurements of birth weight (gm), crown-heel length (cm), and head circumference (cm) at 24 hours of age were compared with the intrauterine growth standards of Usher and McLean.[11]

At a postnatal age of 3 days electroencephalographic recordings of 90 to 120 minutes' duration were done immediately after feeding during the daytime between 11:30 and 16:30 hours with the use of C_4/A_1 and C_3/A_2 scalp electrode placements of Grass gold-plated disk electrodes (7 mm diameter). The signals were processed by Grass 7P511 amplifiers with a time constant of 600 msec. Simultaneous polygraphic recordings were made (Grass Model 78 polygraph) of electroencephalogram, eye movement (E_1/A_1 and E_2/A_1 electrode placement), respiration (Hewlett-Packard Respiration Module No. 78202 A/B in conjunction with a Grass 7P1A amplifier), electrocardiogram standard lead II, and electromyogram from mental and submental muscles (Grass 7P511 amplifiers). A "trigger pulse" was recorded on the marker channel. The electroencephalogram signal and trigger pulse were also recorded on a Hewlett-Packard 3960 Instrumental FM Tape Recorder (2.38 cm/sec tape speed) for off-line computer processing (PDP 8-1).

Table II. Characteristics of the infants

Parameter	Infant			
	Control	Alcohol-exposed	Control	Smoke-exposed
Gestational age (wk)				
37	1	4	0	1
38-40	15	12	17	15
41-42	1	1	1	2
Sex				
Female	9	9	9	9
Male	8	8	9	9
Apgar score				
1-min				
0-3	0	4	0	0
4-6	3	6	0	3
7-8	6	6	5	10
9-10	8	1	13	5
5-min				
0-3	0	1	0	0
4-6	1	1	0	1
7-8	2	5	2	3
9-10	14	9	15	13
Complications				
Respiratory problem	1	5	0	2
Small for gestational age	0	6	0	2
Fetal alcohol syndrome	0	5	0	0
Other congenital anomalies	0	2	0	1
Alcohol withdrawal	0	2	0	0

Table III. Weight, length, head circumference, and electroencephalogram power of sleep in infants of alcoholic and smoking mothers compared to their respective control groups

Group	No. of patients	Birth weight (gm)	Length (cm)	Head circumference (cm)	Integrated EEG power (SUM)		
					Quiet sleep	Intermediate sleep	REM sleep
Control	17	3,466 ± 139	51.9 ± 0.7	34.9 ± 0.4	5.68 ± 0.3	5.89 ± 0.3	2.99 ± 0.19
Alcoholic	17	2,615 ± 151*	48.0 ± 0.9*	32.6 ± 0.5*	8.42 ± 0.91*	10.06 ± 1.64†	6.46 ± 0.71*
Control	18	3,587 ± 105	52.3 ± 0.5	35.0 ± 0.3	6.02 ± 1.24	5.92 ± 0.34	3.17 ± 0.75
Smoking	18	2,973 ± 146*	50.0 ± 1.6*	34.2 ± 0.3	5.89 ± 0.36	5.72 ± 0.45	3.09 ± 0.16

*$p < 0.01$.
†$p < 0.05$.

The trigger pulse subdivided the polygraphic record into 12 second epochs which were numbered sequentially and which were scored for sleep state. The following stages of sleep were differentiated: (1) quiet sleep, (2) indeterminate sleep, (3) active or rapid eye movement (REM) sleep. Power spectral analysis of the electroencephalogram during each sleep state was undertaken by means of methods previously described in detail.[12, 13] The groups were compared with the use of one-way analysis of variance and Duncan's multiple-range test where appropriate.

Results

Effects of alcohol. The characteristics of mothers in the alcoholic and control groups are presented in Table I. More mothers in the alcoholic group were older than 30 years; however, the mean difference in maternal age was only 3.5 years and this difference was not significant. There was no difference in racial origin, parity, or pregnancy risk factors. The alcoholic mothers were, on the average, heavier smokers than mothers in the control group (more than one pack per day). The control group had more mothers from a higher socioeconomic class. Infants delivered of alcoholic mothers were not different from those in the control group with regard to gestational age or sex ratio (Table II); however, these babies did have a lower Apgar score at 1 minute than control infants, but this difference was not significant by 5 minutes of age. The behavior of the alcohol-exposed babies was noted by irritability, tremors, or jittery movements, and two of them had seizures considered to be secondary to alcohol withdrawal. All infants of alcoholic mothers had some complication after delivery: Five infants had respiratory problems

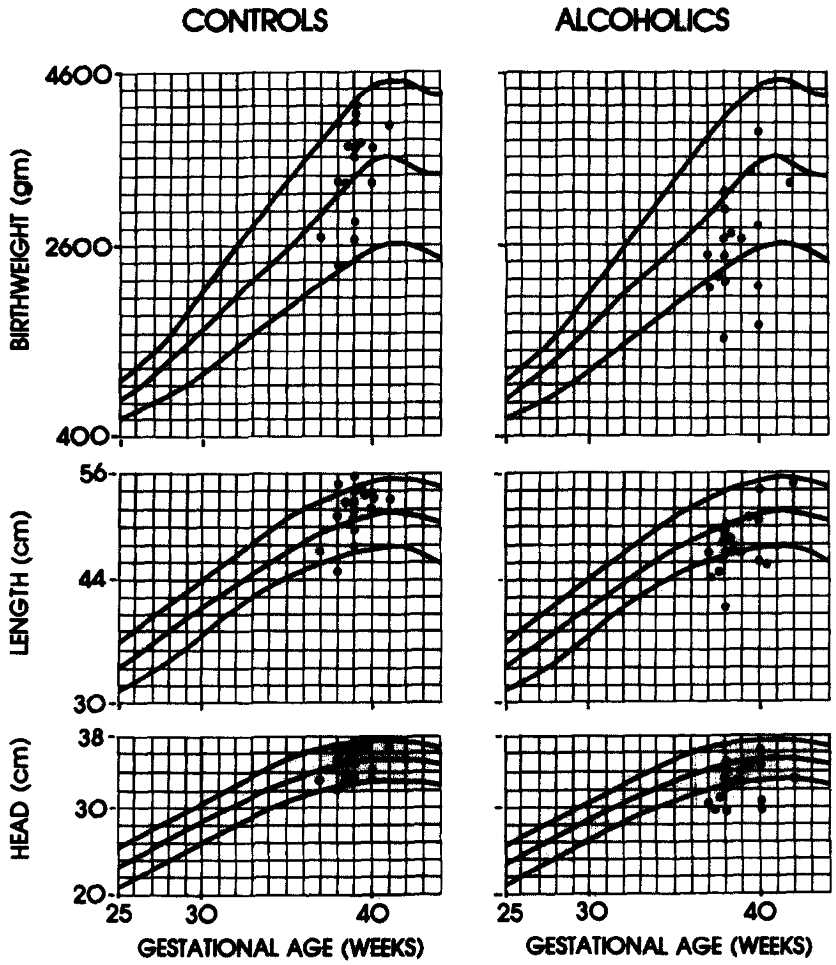

Fig. 1. Comparison of the weight, length, and head circumference in individual infants of alcoholic mothers and control infants.

(three transient tachypnea of the newborn and two pneumonia), six were small for dates, seven had fetal alcohol syndrome or other congenital abnormalities. Only one infant in the control group had transient tachypnea. Infants of alcoholic mothers weighed 850 gm less at birth than the matched control babies, a difference which was highly significant (Table III; Fig. 1). In addition, they were significantly shorter and had smaller head circumferences than control infants.

Visual inspection of the poligraphic record revealed a higher amplitude of electrical activity of the brain in infants of alcoholic mothers (Fig. 2). This increase in amplitude was in several instances of such magnitude that amplification had to be reduced by half. These differences between the alcohol-exposed and control group became more prominent when the sleep electroencephalogram power spectrum was compared. The average integrated electroencephalogram power (SUM) of infants of alcoholic mothers was significantly higher in all three stages of sleep (Table III). The greatest difference in the integrated electroencephalogram power (SUM) was found during REM sleep (Fig. 3). Only two infants in the alcohol-exposed group had an integrated electroencephalogram power during REM which fell within 2 SD of the control group. Infants of alcoholic mothers showed significantly higher electroencephalogram power of most frequency bands (0.1 to 25 Hz) in all three stages of sleep when compared with the control group (Table IV).

Effects of smoking. Characteristics of mothers in the smoking and matched control groups are presented in Table I. There were no differences between the groups in maternal age, racial origin, pregnancy risk factors, or socioeconomic status. Parity in the smoking group was slightly higher than that in the control group but the difference was not significant. None of the control women smoked during pregnancy, whereas smoking mothers all used one or more packages of cigarettes a day.

Infants of smoking mothers had the same gestational

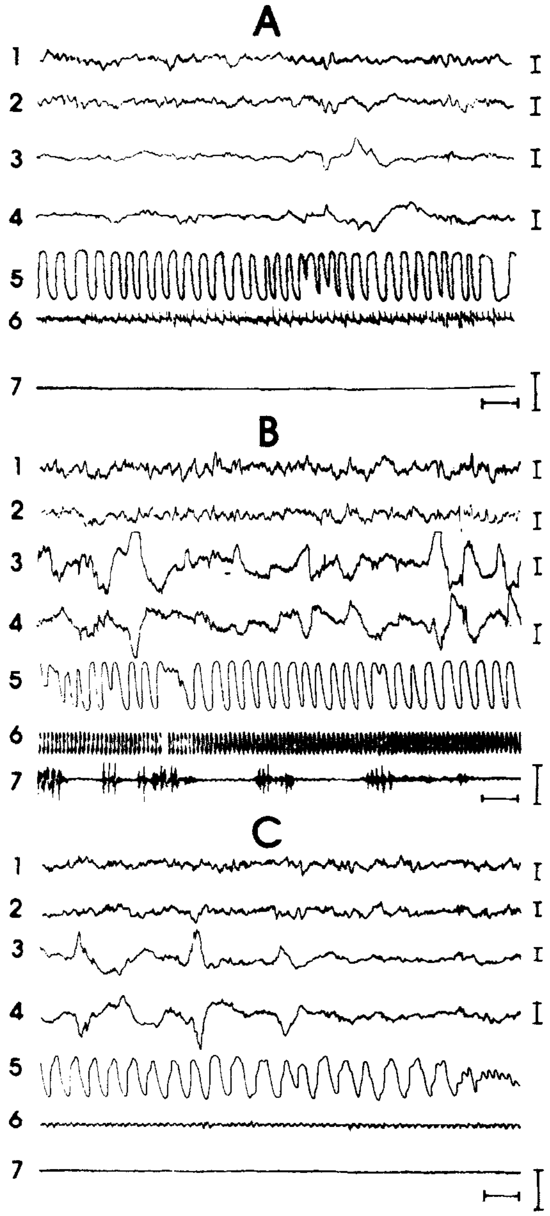

Fig. 2. Polygraphic tracing in term infants (39 weeks' postconceptional age) during REM sleep. *A*, healthy control infant: *B*, infant of alcoholic mother: *C*, infant of smoking mother. *1*, C_3/A_2 electroencephalogram tracing; *2*, C_4/A_1 electroencephalogram tracing; *3*, E_1/A_1 electrooculogram; *4*, E_2/A_1 electrooculogram; *5*, respiration; *6*, electrocardiogram (II standard lead); *7*, electromyogram from submental muscles. Calibration: horizontal = 3 seconds; vertical = 100 mv.

age and sex ratio and Apgar score at 5 minutes as those of control mothers (Table II). The Apgar score at 1 minute was slightly but significantly lower in the infants of smoking mothers. Infants of smoking mothers showed significantly lower birth weights and lengths in comparison with control infants (Table III). However, head circumference was not affected.

Infants of smoking mothers and control mothers

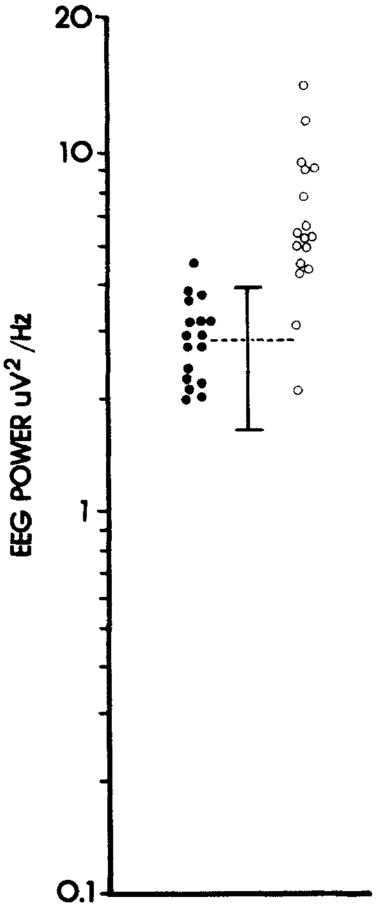

Fig. 3. Total electroencephalogram power (SUM) (mv^2/Hz) in control infants *(closed circles)* and infants of alcoholic mothers *(open circles)* during REM sleep. Mean power ± 2 SD is shown for the control infants. Note that only two infants of alcoholic mothers fall within 2 SD of the control group.

showed the same behavioral and electroencephalogram pattern (Fig. 2, *C*), and there were no significant differences in either power spectra value of individual frequency bands of SUM in quiet, indeterminate, or REM sleep between the two groups (Table III).

Comment

During the last decade evidence has become available which indicates that alcohol may have morphologic, neurological, and behavioral teratogenic effects on the developing fetus. In 1968, Lemoine and associates[1] from France were the first to describe dysmorphologic signs in infants of alcoholic mothers and which was a few years later termed by Jones and Smith,[3] the "fetal alcohol syndrome." This syndrome includes prenatal and postnatal growth retardation, microencephaly, an unusual facial pattern typified by short palpebral fissures, maxillary hypoplasia, flattened nasal bridge, thin upper lip, ear anomalies, minor joint and limb anomalies, and cardiac defects. In some instances, se-

Table IV. Statistical differences in EEG (C_3/A_1) power of frequency bands and SUM between group infants of alcoholic mother and group infants of healthy mother in quiet, indeterminate, and REM sleep

	Sleep state		
Frequency band (Hz)	*Quiet*	*Indeterminate*	*REM*
Delta 1 (0.10-1.48)			*
Delta 2 (1.56-3.51)	†	†	*
Theta 1 (3.61-5.57)	*	*	*
Theta 2 (5.66-7.52)	*	*	*
Alpha 1 (7.62-9.47)	*	*	*
Alpha 2 (9.57-12.50)	*	†	*
Beta 1 (12.50-17.48)	†		†
Beta 2 (17.58-25.0)			
SUM (1.56-25.0)	*	†	*

$^{*}p < 0.01$ For differences when EEG power of infants of alcoholic mother was higher than infants of healthy mother.
$^{\dagger}p < 0.05$.

vere brain malformations were described.[14] Dysmorphologic effects of ethanol are often associated with functional and behavioral teratogenic effects such as increased irritability, tremulousness, jitteriness, seizures, motor incoordination, weak sucking, and motor and mental developmental delay with borderline to retarded intelligence.[14, 15] Since neuronal multiplication in the central nervous system predominantly takes place during the second trimester, this may be the critical period during which the fetal exposure to alcohol results in the microencephaly with a smaller number of neurons, neuronal and glial dysplasias, and heteroptic cell clusters.[14, 16] Alcohol exposure during the third trimester may result in selective neocortical and cerebellar defects since this is the critical time for development of those brain structures.

Many investigators have reported effects of maternal smoking on reproductive function, growth, and development of offspring. There is an increased incidence of bleeding during pregnancy, abruptio placentae, placentia previa, prolonged rupture of membranes, spontaneous abortion, and prematurity.[17] Although investigators generally have reported no relationship between smoking and congenital abnormalities, there are some reports of increasing incidence of anencephaly, cleft palate and harelip, and congenital heart defects.[8, 18] Simpson and Linda[19] first identified a decrease in birth weight of infants of smoking mothers and reported that prematurity rate increased with the number of cigarettes smoked per day. The association between smoking and intrauterine growth retardation has subsequently been described by many other investigators.[1, 5, 6, 17]

We have previously described a significant increase in the electroencephalogram power spectrum in infants of alcoholic mothers.[4] Because most alcoholic mothers that we have studied were also heavy smokers, it has not been possible to separate the effects of smoking on the fetus from those of alcohol. The present study was undertaken in order to separate the confounding factor of smoking in alcoholic mothers.

The present study indicates the effects of maternal alcohol consumption on growth and on the brain electrical activity of the offspring. Measurements of the weight, length, and head circumference at birth of groups of infants from alcoholic mothers were significantly reduced in comparison with a carefully matched control group. Some of the alcohol-exposed infants also showed congenital and dysmorphologic changes described as fetal alcohol syndrome. This dysmorphologic effect of alcohol was also associated with increased irritability, tremulousness, jitteriness, and seizures in the group of infants of alcoholic mothers. Fifteen of 17 infants of alcoholic mothers showed generalized electroencephalogram hypersynchrony, seen in all stages of sleep. Hypersynchrony was also observed on visual inspection, but precise quantitation was achieved by computerized electroencephalogram spectral analysis. The power of most frequency bands was significantly higher in all three stages of sleep. The average integrated electroencephalogram power (SUM) was significantly increased in quiet sleep, indeterminate sleep, and REM sleep, and this change was 148% to 212% of the control values. The hypersynchrony was most pronounced in REM sleep.

Infants of smoking mothers also had a decrease in birth weight and length when compared with a closely matched control group and we could not differentiate an effect of alcohol on fetal growth from that of smoking. These data are consistent with the data of van den Berg[20] who found lower birth weights in infants of drinking, smoking mothers compared with infants of nonsmoking drinking mothers. Interestingly, infants of alcoholic mothers had smaller head circumferences than control infants, but infants of smoking mothers had the same head circumferences as control infants. These data are consistent with the notion of an adverse effect of alcohol on fetal brain development. The difference in head circumference between infants of heavy drinking mothers and their matched control group was only 2.3 cm. However, with the use of the determined relationship between head circumference and brain volume,[21] this difference represents a decrease in brain volume of 92 ml in the infants of alcoholic mothers.

Our study clearly indicates that maternal smoking is not related to the electroencephalogram hypersynchrony found in infants of alcoholic mothers, since the electroencephalogram power spectral analysis was not different in infants of smoking and nonsmoking mothers in any sleep state. Thus, the electroencephalographic changes in infants of alcoholic mothers clearly are unrelated to the morphometric effects of adverse prenatal factors such as smoking and seem to be a specific effect of alcohol. We would also suggest that the current definition of fetal alcohol syndrome based on morphologic criteria may be inadequate. Our findings indicate that heavy ingestion of alcohol by the mother affects the neonatal electroencephalogram even in the absence of dysmorphology. Thus, the electroencephalogram may be the most sensitive indicator of fetal alcohol toxicity.

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